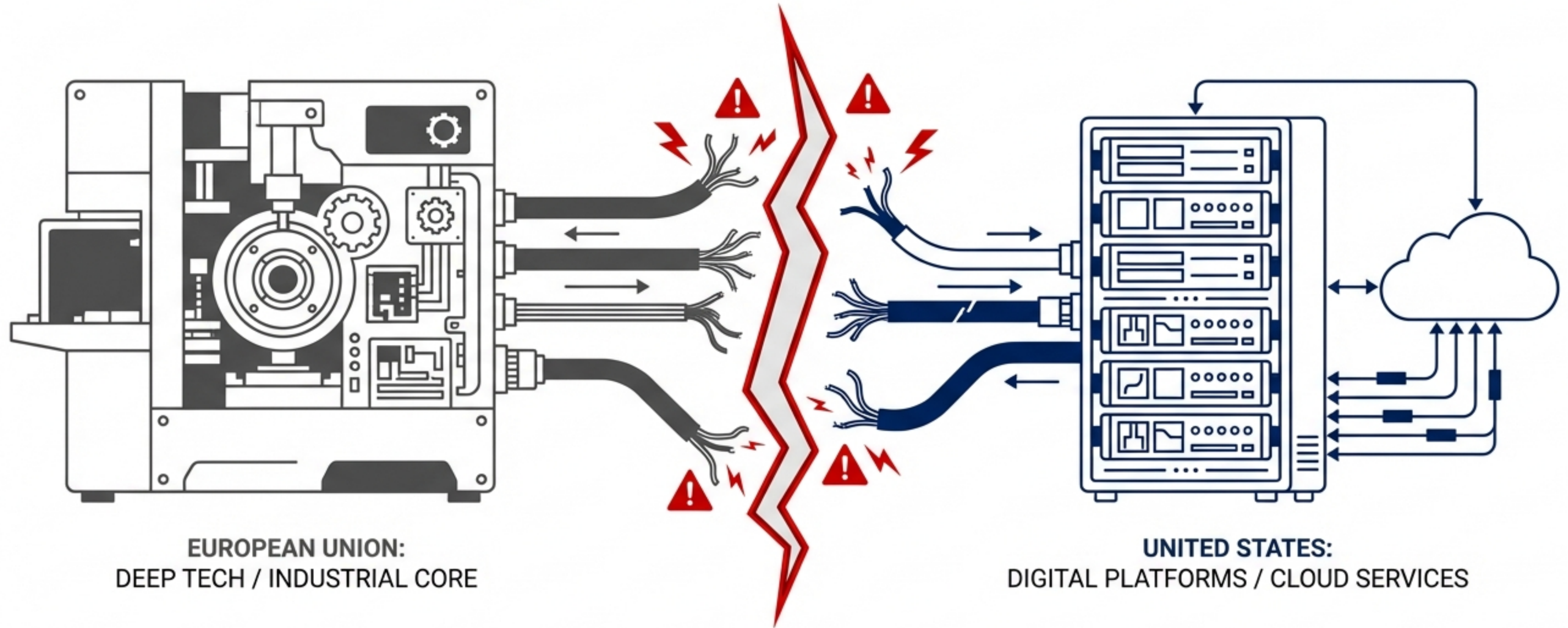


The Transatlantic Tech Divorce: A Scenario Analysis of Decoupling

Assessing the economic and operational impact of a hypothetical rupture between the European Union and United States technology sectors.

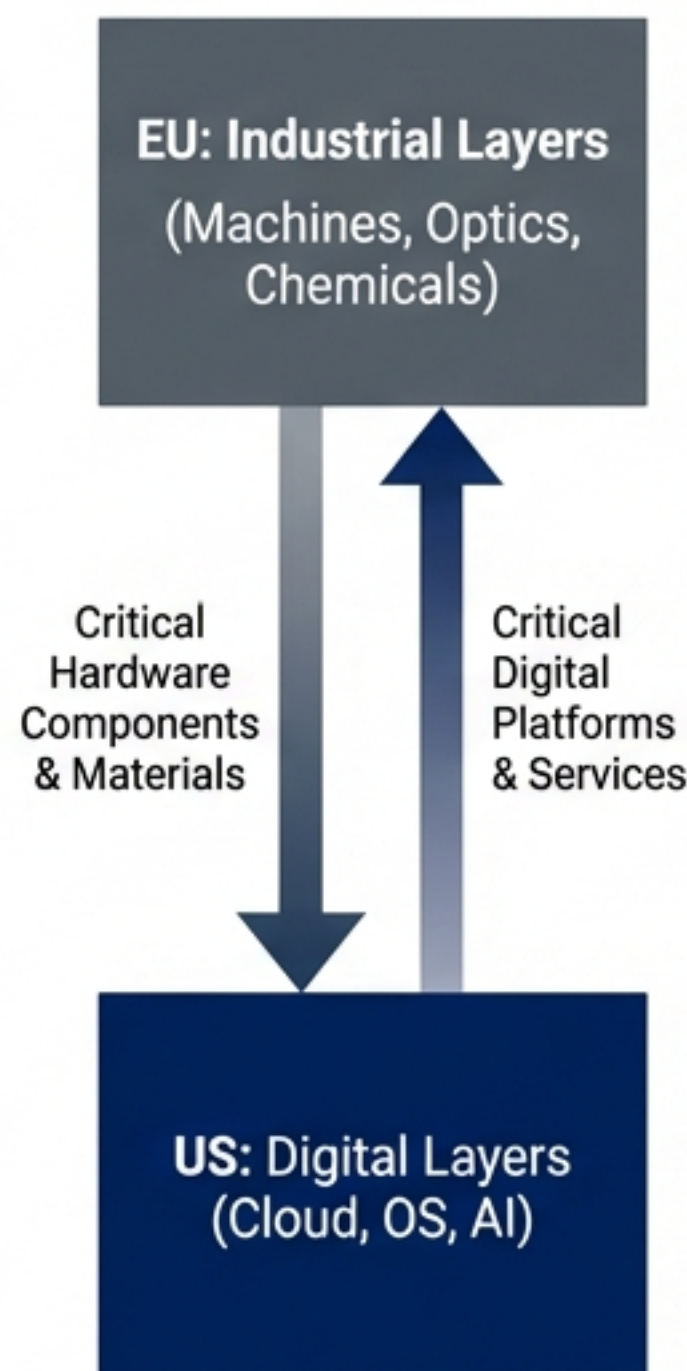


Source Data: 'What if Europe Severed Tech Ties with the U.S.?' Scenario Assessment.

Executive Summary: A Roadmap to Mutually Assured Destruction

The Core Thesis: Asymmetric Interdependence

The relationship is defined by a mismatch in leverage. Europe controls the 'invisible' industrial production layers (machines, optics, chemicals). The US controls the 'visible' digital consumption layers (cloud, OS, AI).



The Scenarios



Scenario A: Supply Shock (EU Blocks Exports)

Result: US chip fabrication halts (ASML dependency) and industrial design stalls (Dassault/SAP).

US Chip Fabrication Halts (ASML)

Industrial Design Stalls (Dassault/SAP)



Scenario B: Demand Shock (EU Blocks Imports)

Result: 'Tech paralysis' in European infrastructure and massive revenue loss for Silicon Valley (20–30% drop).

European Infrastructure Tech Paralysis

Silicon Valley Revenue Loss (20–30% Drop)

The Bottom Line

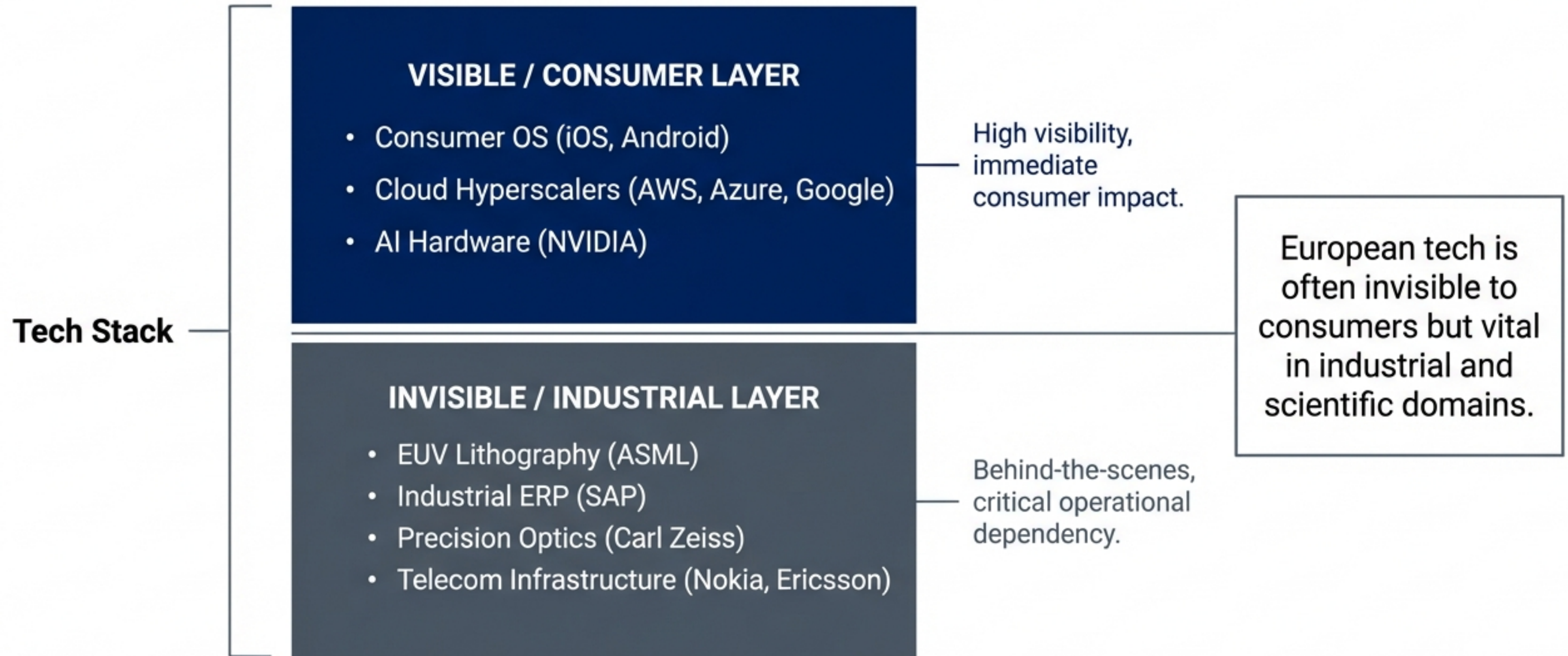
A full decoupling is a nightmare scenario for European functionality and a financial disaster for US markets. The only geopolitical winner in a fractured Western ecosystem is the 'Third Player': China.

- Full Decoupling = Nightmare for EU Functionality
- Full Decoupling = Financial Disaster for US Markets
- Only Geopolitical Winner = 'Third Player': China



FRACTURED ECOSYSTEM

The Balance of Power: Visible Platforms vs. Invisible Machinery

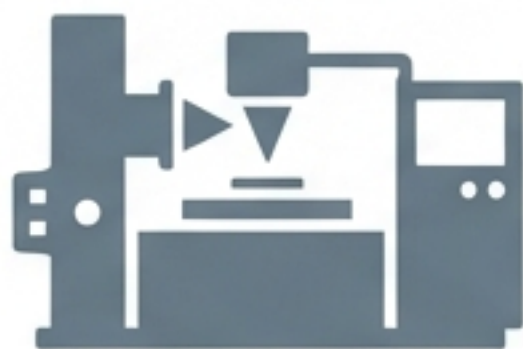


The Semiconductor Chokepoint: Europe's 'Big Hammer'

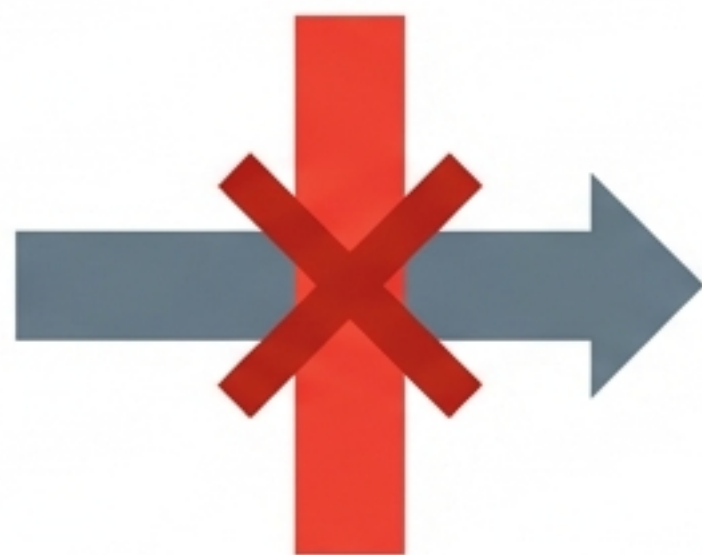


100%

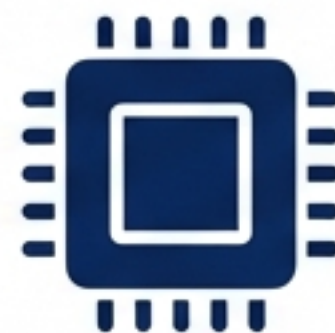
Global market share of
Extreme Ultraviolet (EUV)
lithography held by ASML.



ASML
(Netherlands)



Global Fabs
(Intel, TSMC Arizona)



**Finished
Semiconductors**

Impact on US

- **Immediate:** Without spare parts and calibration software, US chip output falls within weeks.
- **Strategic:** Bottlenecks production of military chips and data centre GPUs.

The Reality Check

Even with the US CHIPS Act investment, the machinery required to run domestic fabs remains a European monopoly.

The Industrial Backbone: How European Code Runs American Companies

Enterprise Software



SAP (Germany)

Runs the logistics and financial back-end for major US corporations. Halting updates would paralyze supply chains.

Aerospace Design



Dassault Systèmes (France)

Case Study: Boeing relies on CATIA software for aircraft design. License revocation disrupts engineering workflows instantly.

Telecom Infrastructure



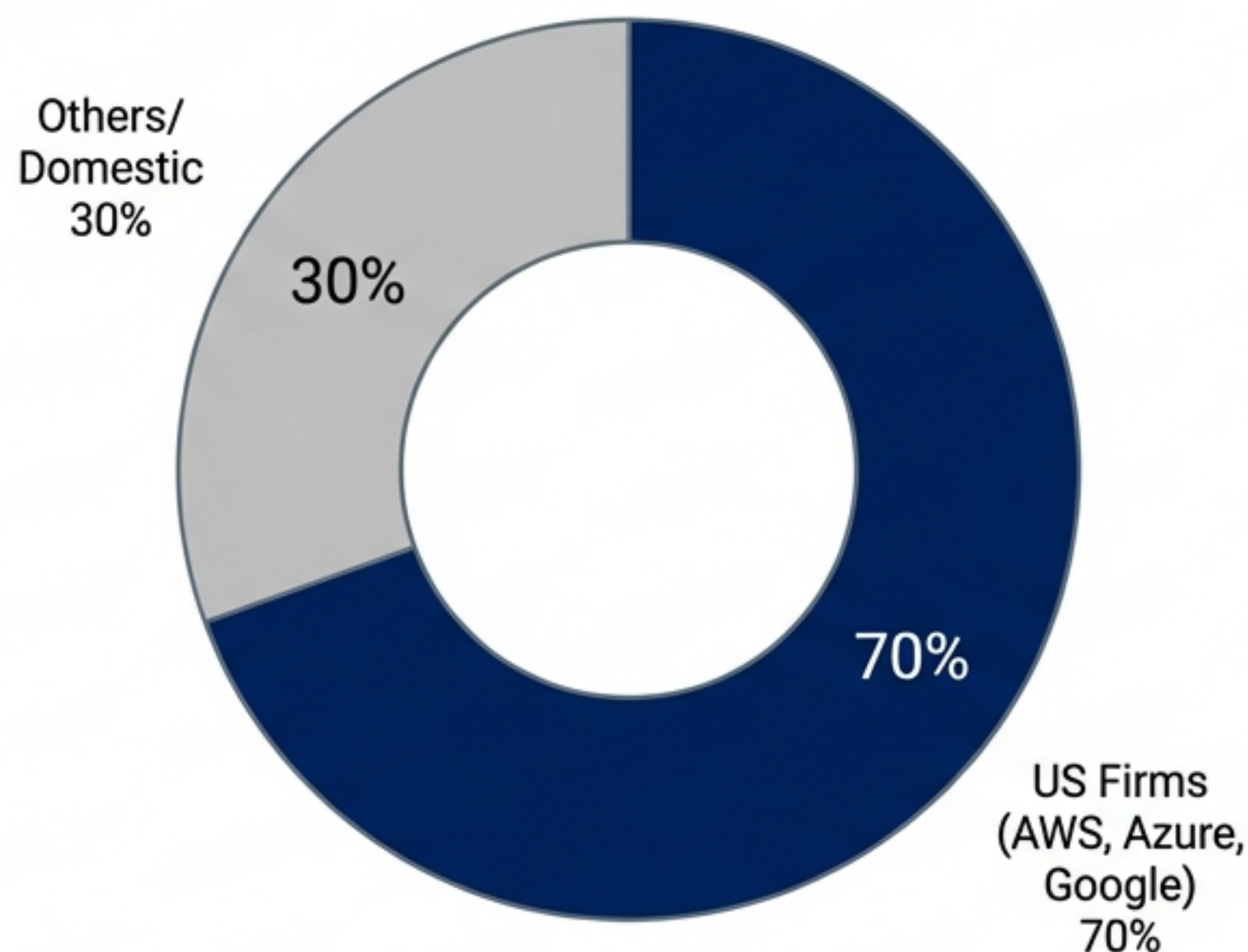
Nokia (Finland) & Ericsson (Sweden)

A large share of US 4G/5G infrastructure is Nordic. "Rip and replace" costs are prohibitive; network rollouts would stall.

Conclusion: While the US can source alternatives eventually, short-term supply chain disruptions would be **enormously disruptive**.

Cloud Collapse: The Cost of Digital Sovereignty

Control of European Cloud Infrastructure



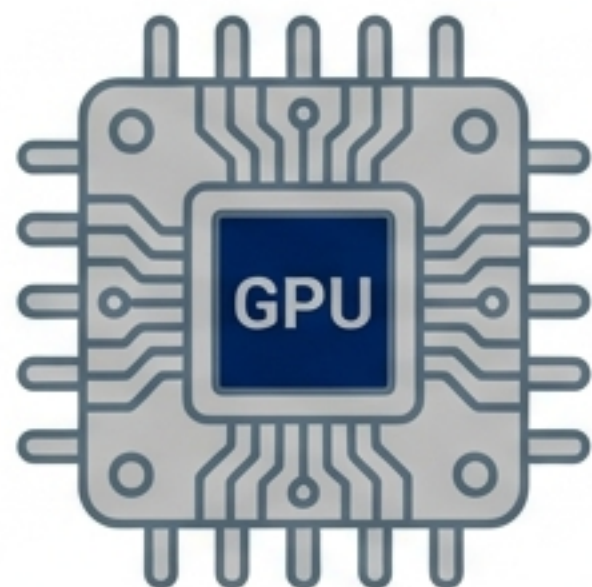
The Paralysis Scenario (Germany Case Study)

- **Capacity Shortfall:** Immediate loss of 1,200 MW of IT capacity (approx. 40% of total).
- **Data Deficit:** 40 exabytes of storage inaccessible immediately.
- **HPC Impact:** 2/3rds of German high-performance computing runs on US cloud.

Immediate Chaos: Outages in energy grids, healthcare systems, and government services. Minimum recovery time: 18–24 months.

The AI Deficit: Cut Off from the Bleeding Edge

Hardware Dependency



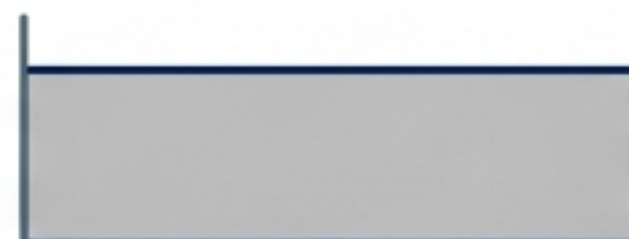
85%

of AI-critical GPUs in Europe are provided by US firms (e.g., NVIDIA).

Consequence: European AI labs **frozen** on legacy hardware. New "AI diffusion rules" already demonstrate US willingness to **limit chip exports**.

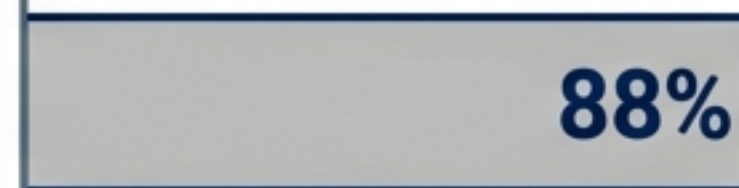
Software Gap

EU Listed Companies



rely on US cloud/email

UK Listed Companies

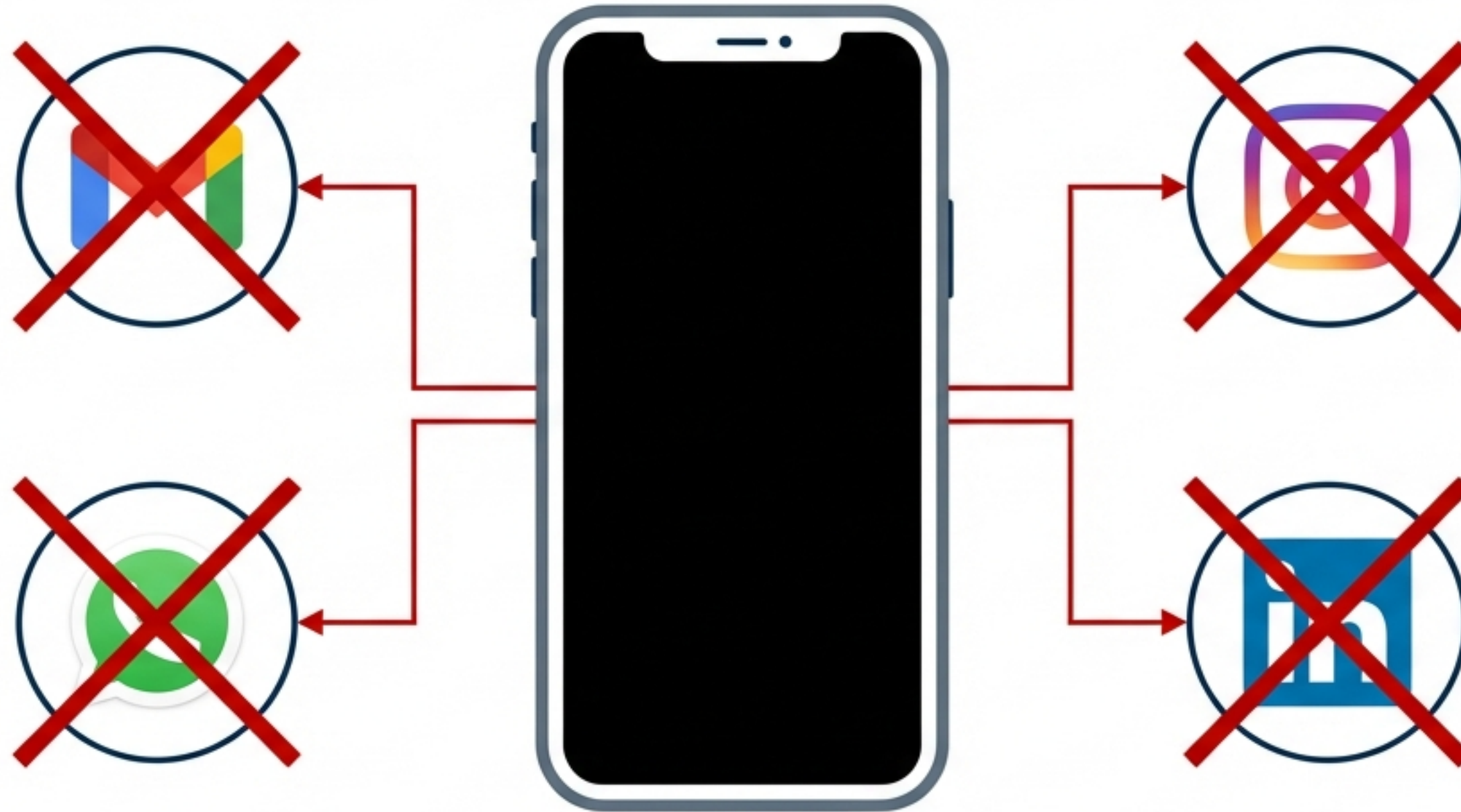


rely on US cloud/email

Consequence: High dependency on US software infrastructure creates vulnerability. A sudden decoupling would disrupt essential services and data access.

Analysis: Initiatives like the European Chips Act are long-term plays. They cannot prevent an **immediate performance lag** in a sudden decoupling.

The Consumer Shock: A Society Without an OS



The Mobile Duopoly

Google (Android) and Apple (iOS) hold effectively 100% of the mobile OS market in Europe.

The Logistical Nightmare

Replacing operating systems for hundreds of millions of devices is a Herculean task due to the absence of a mature 'Euro-OS'.

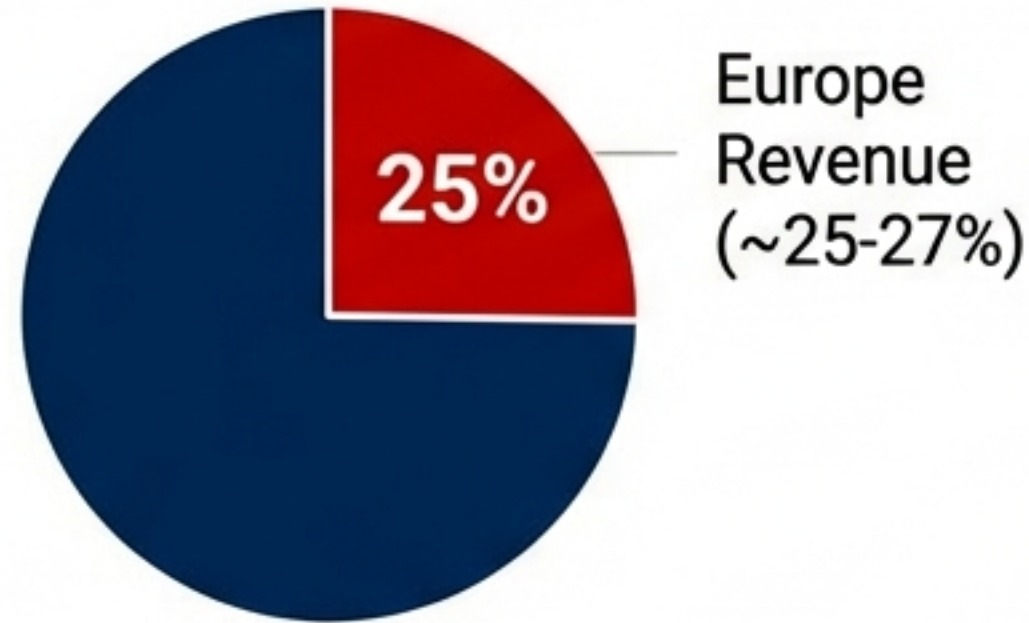
Data Sovereignty Risk

Migrating petabytes of user data from US-owned data centres risks massive bottlenecks and data integrity loss.

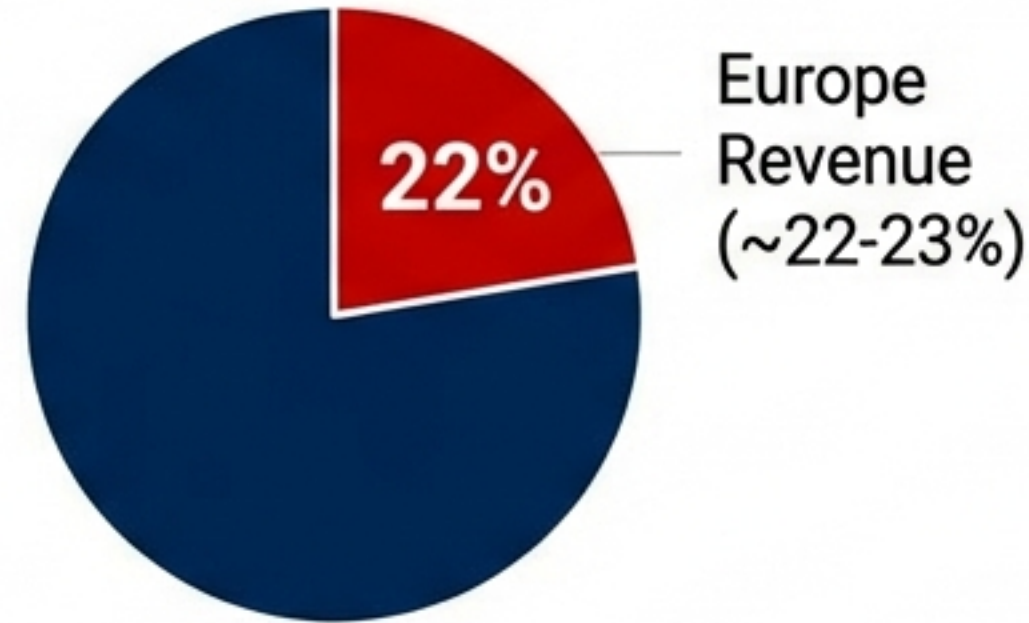
Impact: Massively disruptive to daily communication, commerce, and social stability.

The Financial Blast Radius: US Revenue Exposure

Apple



Meta



The Trade Bazooka

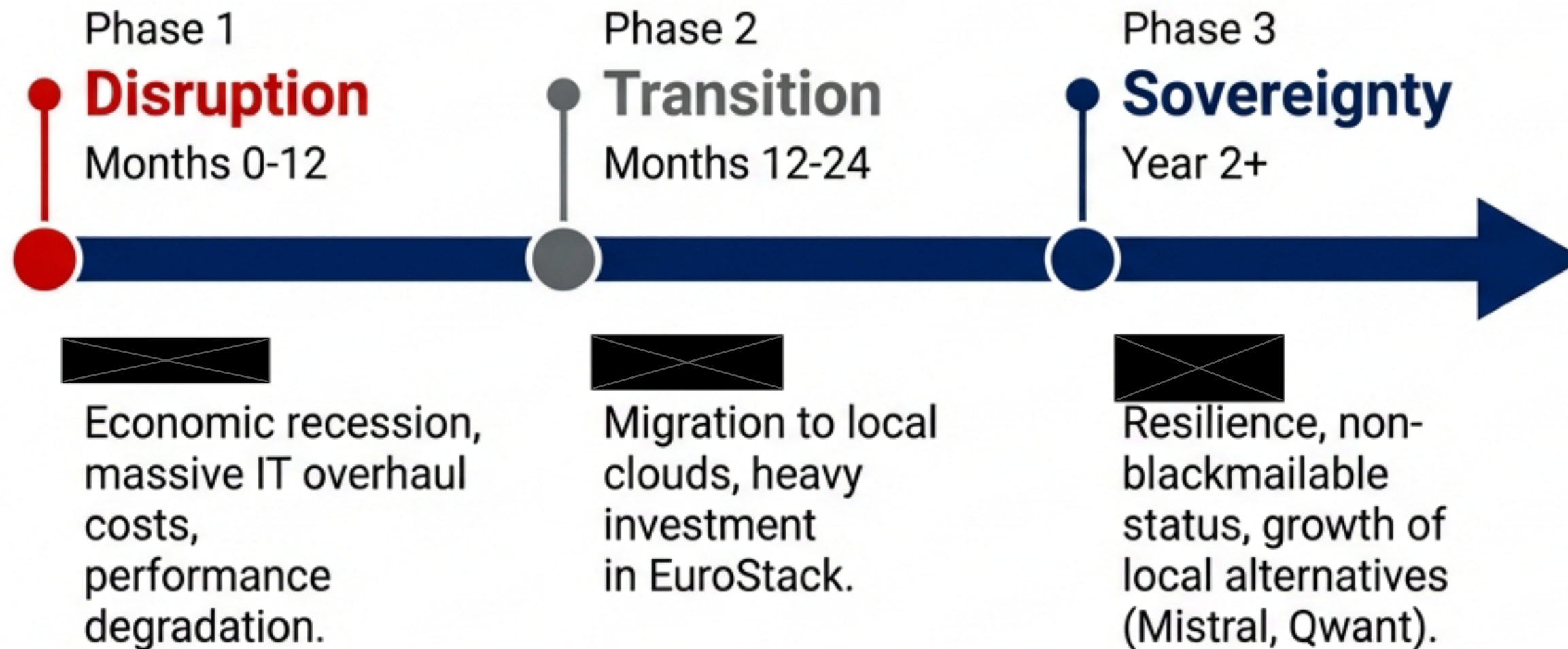
Blocking US companies from the EU Single Market is a financial weapon of mass destruction.

Aggregated Impact: US tech giants face an instant revenue loss of ~30%.

Secondary Effects

- **Stock Valuation:** Immediate plummeting of share prices.
- **R&D Budgets:** Global reduction in innovation spend due to lost scale.
- **Talent Access:** Loss of access to European open-source contributions (e.g., Linux roots) and engineering talent.

The Operational Blast Radius: Short-Term Chaos, Long-Term Autonomy

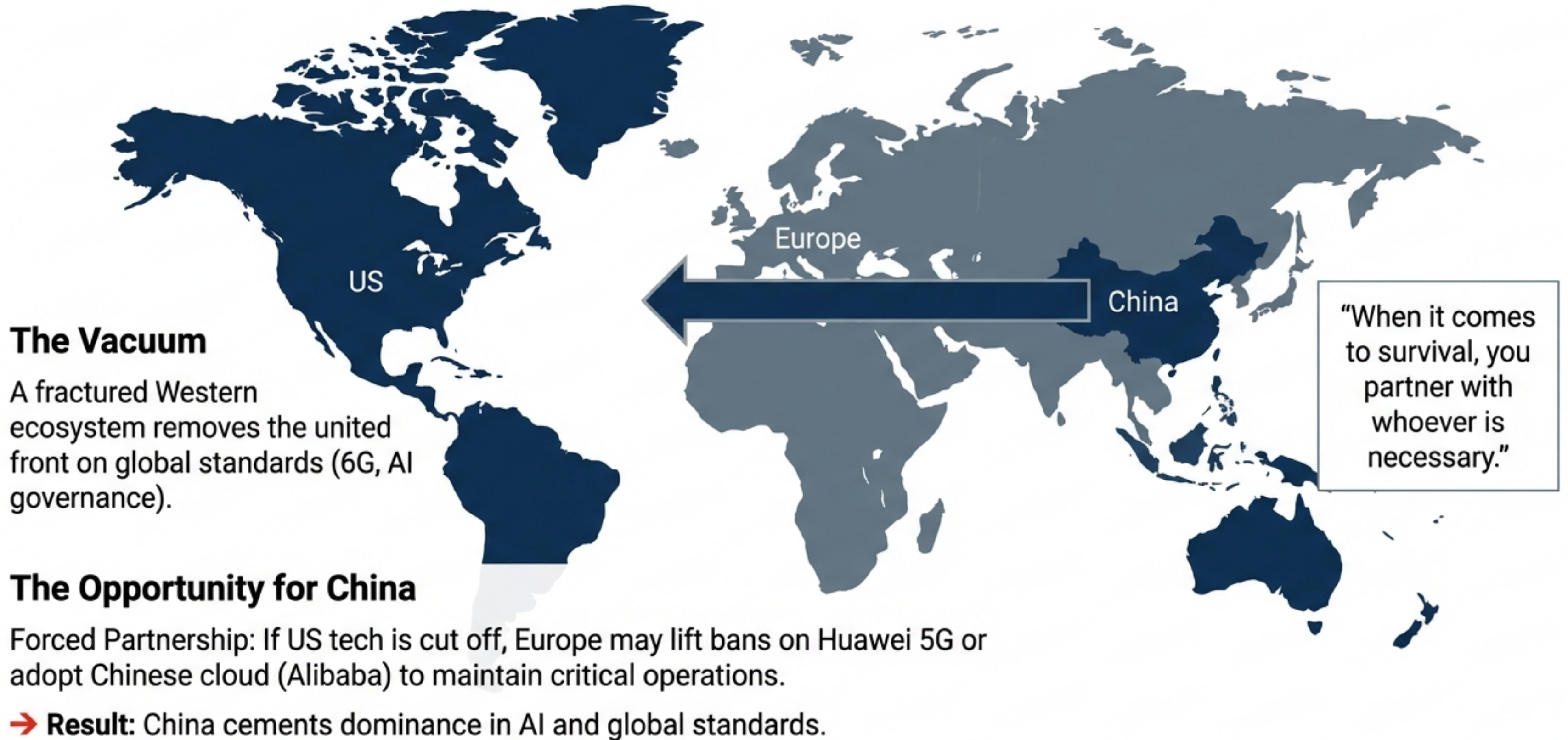


“If you can’t innovate, you regulate – but now Europe must innovate.”

Strategic Outcome

Europe trades short-term efficiency for long-term resilience and geopolitical independence.

The Geopolitical Winner: The Third Player Advantage



Inter Tight: The Verdict: Mutually Assured Destruction

US Losses	EU Losses
• Loss of wealthiest export market. • Revenue Impact: 25-30% drop for Tech Giants. • Loss of global standards leadership. • Reduced Soft Power and Intelligence access.	• Loss of tools required to run a modern economy. • Operational instability and service outages. • Loss of access to bleeding-edge AI. • Immediate Consumer inconvenience.

**Decoupling is akin to 'shooting both sides in the feet.'
The US loses growth; Europe loses functionality.**

Strategic Imperative: 'Just Enough' Sovereignty

